FEATURES OF THE MOST STREAMED SONGS OF 2024

#DATABEASTS

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NEXT

INTRODUCTION

An agency dedicated to helping music professionals, artists, and analysts understand the key trends behind the most streamed songs on Spotify by leveraging data-driven insights. We would like to highlight the trends and characteristics of the most streamed songs in 2024.

Our intention: discover patterns and characteristics that define hit songs, turning data into meaningful, useful insight.

Goal: empower artists, producers, and industry professionals to make informed creative and strategic decisions in the music industry

DATASETS

The dataset is created by a comprensive merge of four distinct tables that allows us to have a full overview about the most streamed spotify songs in 2024 and their characteristics.
We started using the Spotify's Most Streamed Songs 2024 and we combined it with the following datasets: Spotify songs, Spotify Music Dataset, Top Spotify Songs in 73 Countries.
That procedure allows us to obtain a list of all the interested songs with their features.
The first dataset contains features such as: song names, artist names, total streams and daily stream counts.
The other three datasets used in this project provide information on song popularity and audio features such as energy, danceability, loudness, liveness, and valence, covering globally popular tracks.

INSIGHTS FOR ARTISTS AND LABLES

1 Most popular and streamed artists and album

Objective:

- identify the artists and albums that consistently dominate the music charts based on their popularity and streaming performance
- examine the correlation between streaming numbers and overall popularity

INSIGHTS FOR ARTISTS AND LABLES

2

Audio characteristic of the most listened songs

Objective:

- visualize how danceability, energy, acousticness are distributed over popular songs
- find the audio features combinations that lead to a popular song
- see which bpm levels are the most used

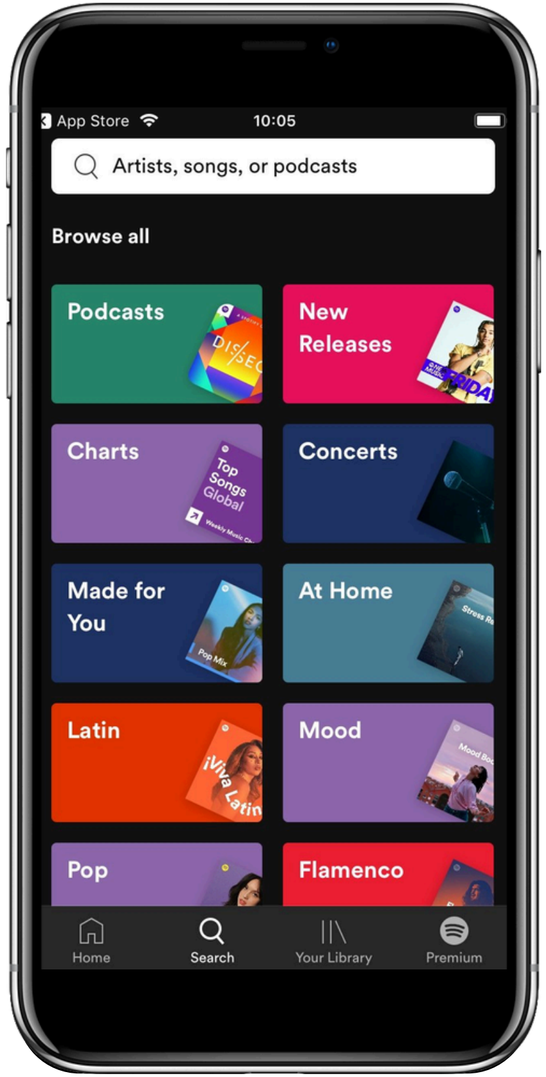
INSIGHTS FOR ARTISTS AND LABLES

3 Correlation between duration and success

Objective:

- analyse whether the most popular songs tend to be shorter or longer
- examining the relationship between song duration and popularity to determine if there is a preference for concise or if longer compositions

FONTS AND PALETTE



Aa

sans-serif fonts like Roboto for clarity and modern appeal.
We highlight key figures or trends with larger or bold fonts to guide the viewer's focus. Keep text minimal on the charts, use tooltips or legends for detailed explanations.

1 TOP 10 MOST STREAMED SONGS AND THEIR ARTISTS

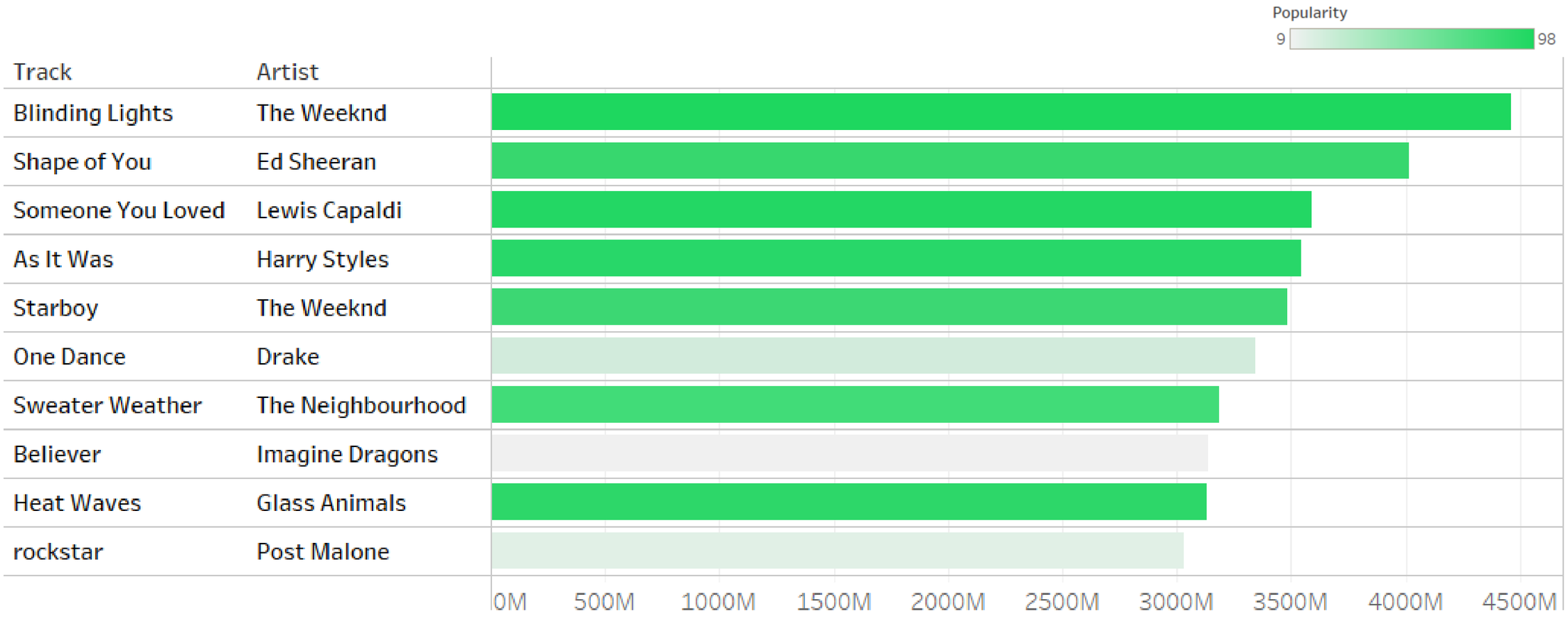


Chart Description:

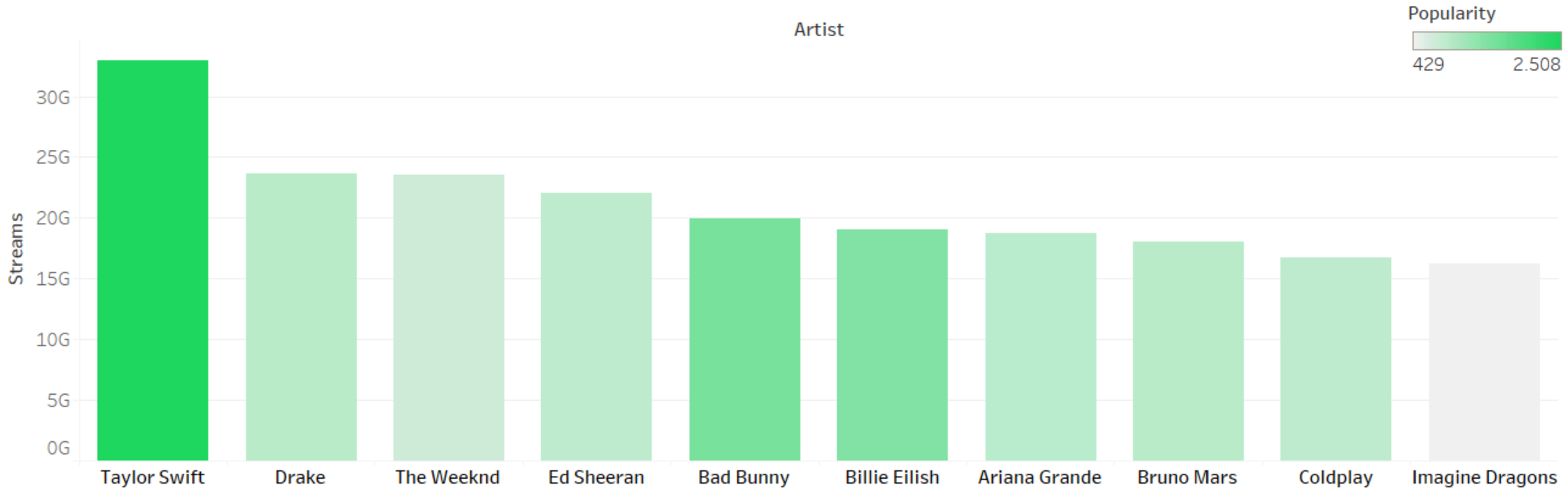
The chart shows the top 10 most streamed songs, along with their respective artists. The length of the bars represents the total number of streams each song has accumulated, with a green color scale indicating its popularity level (from 9 to 98), darker green shades represent the most popular songs.
The top songs include:

- "Blinding Lights" - The Weeknd, with the most streams, exceeding 4.5 billion streams.
- "Shape of You" - Ed Sheeran, follows in second place, with a slightly lower count.
- "Someone You Loved" - Lewis Capaldi, in third place.

Analysis and conclusions:

The Weeknd is the only artist with two songs on the list ("Blinding Lights" and "Starboy"), indicating his great musical influence, apart from that there’s a certain diversification of artists in the chart.
The popularity of the songs is generally high, but as we can see there are some tracks with low popularity. Probably that songs has accumulated lot of streams during the year but they are not still so popular.
All the top 10 most streamed songs have a value above 3 billion streams, remarking also the popularity of the Spotify music platform which is worldwide used.

1 TOP 10 MOST STREAMED ARTISTS



Songs count:	33	21	11	12	22	18	15	12	11	10

Chart Description:

The bar charts represents the total number of streams received by the most streamed artists on Spotify.
Taylor Swift is the artist with the most streams, followed by Drake, The Weeknd and Ed Sheeran.
The bottom row shows the number of songs for each artists.

Analysis and conclusions:

- The chart shows the success of Taylor Swift, who is the number one artist for streams. As we can see she has an high number of songs, 33 hits, and that can confirm her dominance on Spotify.
- Other artists such as Bad Bunny, Drake and Billie Eilish have an high number of hits suggesting that they are not only widely listened to, but also have a large catalog.
- Some artists, such as Ed Sheeran and The Weeknd, have a high number of streams but fewer songs than others, 11 for The weekend and 12 for Ed Sheeran, indicating that their individual songs may have a greater impact in terms of streams. This is confirmed by looking at the previous "top 10 most streamed songs and their artists" graph chart as there are two songs by The Weeknd (“Blinding Lights” and “Starboy”) and one by Ed Sheeran (“Shape of you”).
- The correlation between number of songs and streams is not direct: some artists get a lot of plays with a small number of songs, while others accumulate many streams thanks to a larger catalog. The overall popularity of an artist therefore does not depend only on the number of streams.

1 RELATION BETWEEN POPULARITY AND STREAMS

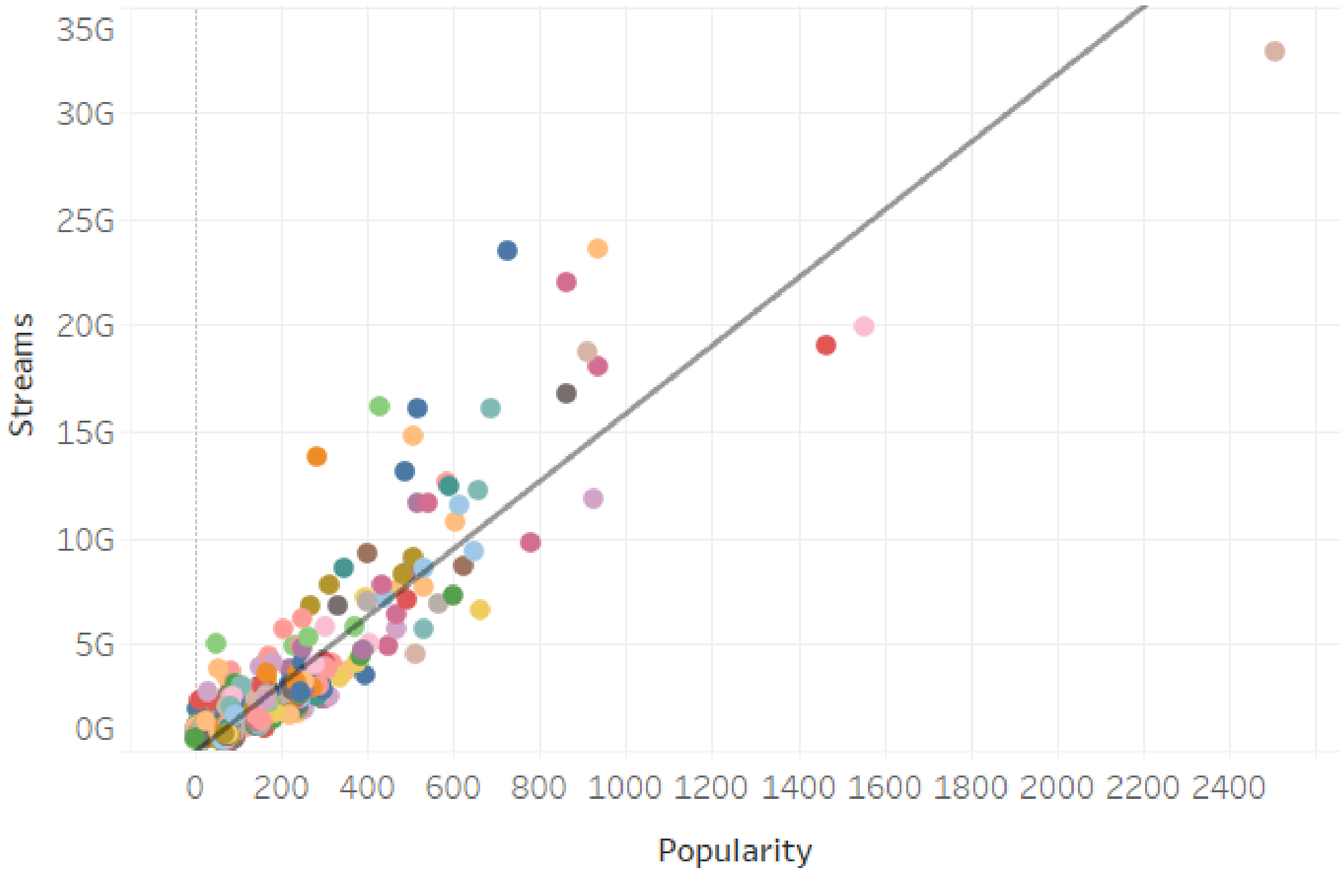


Chart description:

The graph is a scatter plot that represents the relationship between the popularity of a song (horizontal axis) and the number of streams received (vertical axis). Each point in the graph represents a song, and the trend line highlights the positive correlation between the two parameters.

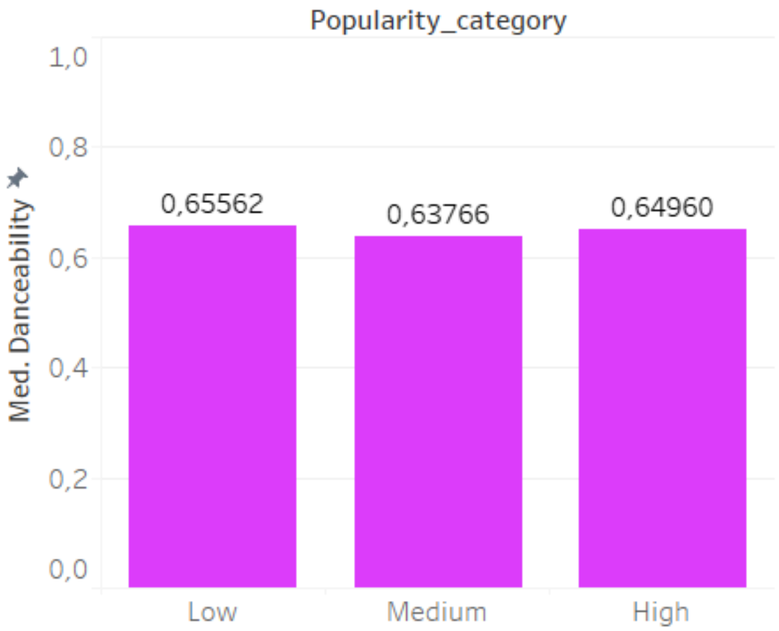
Analysis and Conclusions:

- A positive correlation is observed between popularity and number of streams: songs with an higher popularity score tend to have more streams.
- However, some points deviate from the general trend, indicating that there are songs with many streams but relatively low popularity, and vice versa. This could be due to factors such as user engagement, playlists they are included in, or temporary virality.
- Most of the songs are concentrated in the lower left part of the chart, but there are a few isolated points in the upper right represent extremely popular artists, these are Billie Elish, Bad Bunny and Taylor Swift.

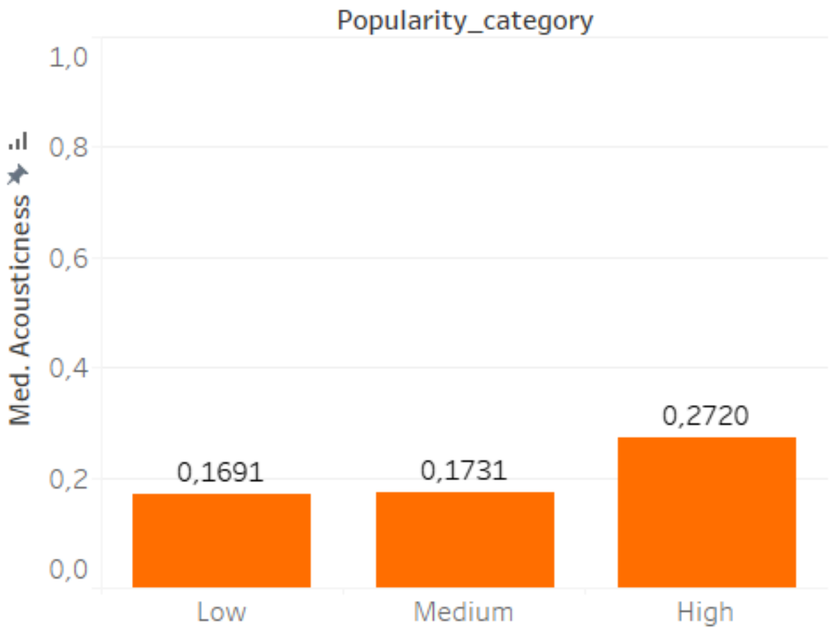
LET'S NOW GO DEEP INTO THE AUDIO FEATURES OF THE MOST LISTENED SONGS

2 DANCEABILITY, ACOUSTICNESS AND ENERGY LEVELS

Danceability level



Acousticness level



Energy level

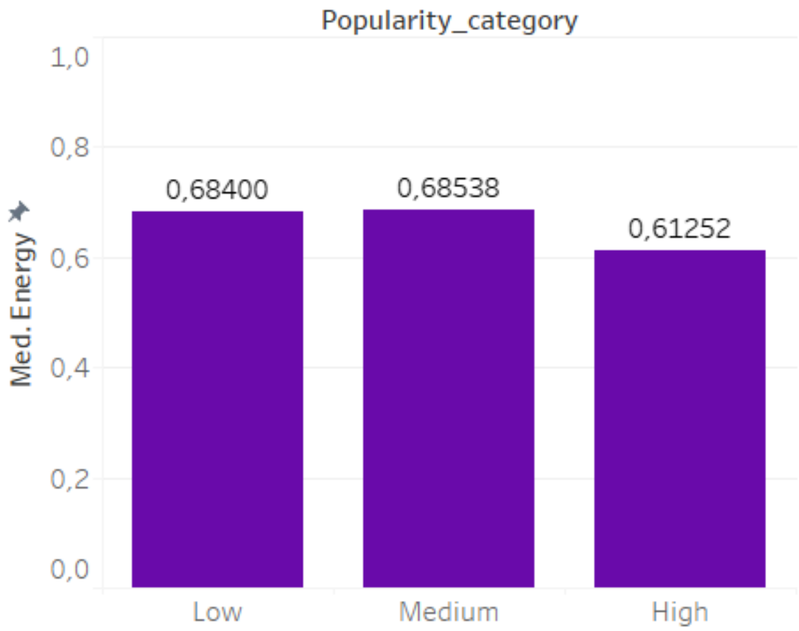


Chart Description:

The chart shows the audio characteristics of the most listened songs on Spotify, dividing the songs into three popularity categories: low (with popularity lower than 71), medium (with popularity between 71 and 78), and high (with popularity higher than 78). This division was based on percentiles, ensuring that each band contains the same number of songs.

The three metrics analyzed are:

- Danceability – Indicates how suitable a song is for dancing, with higher values suggesting greater danceability.
- Acoustics – Represents the probability that a song is acoustic, with higher values indicating a greater acoustic component.
- Energy – Measures the intensity and power of the song, with higher values indicating more energetic songs.

Analysis and conclusions:

- Danceability: The most popular songs (High) have a similar level of danceability to the less popular ones (Low), with values slightly higher than the medium category. This suggests that danceability is not a determining factor for popularity, even if the most listened songs tend to be slightly more danceable. In general we can notice that the most streamed songs have a fairly high level of danceability.
- Acousticness: The most popular songs (High) have a higher acoustic value than the other two categories. This may indicate a trend toward songs with more pronounced acoustic elements among the most played hits. In general we can notice that the most streamed songs have quite low level of acousticness.
- Energy: Less popular songs (Low) and songs of average popularity (Medium) have higher energy levels than the most popular songs (High). We can see that there is no substantial difference in energy levels between the three popularity categories. However, the energy levels between songs are generally quite high.

2 CORRELATION BETWEEN DANCEABILITY, ACOUSTICNESS AND ENERGY

Chart Description

The chart represents the relationship between three audio characteristics of the most listened songs: danceability, acousticness, and energy. The data is divided into intervals for each characteristic and colored based on the popularity of the song:

- Green for high popularity songs
- Yallow for medium popularity songs
- Red for low popularity songs

Each circle shows the average popularity value of the songs belonging to each combination of intervals.

Analysis and Conclusions:

Danceability vs Acousticness

- Songs with Danceability level around 0.3 are really popular, indeed there are many green dots in this range. Popularity is highest when Acousticness is around 0.7, indicating that songs with a high acoustic component tend to be more appreciated.

Acousticness vs Energy

- Again, the most popular songs are found when Acousticness is between 0.5 and 0.9, especially with a value of around 0.7. The most favorable energy level for a hit is between 0.2 and 0.3, suggesting that songs with low-moderate energy work better than those that are too intense.

Danceability vs Energy

- Songs with high energy (over 0.5) tend to have lower popularity. Values between 0 and 0.5 are preferred, in particular 0.2 is the one with more green dots. We can find some green dots in different levels of danceability ([0.2, 0.3] and [0.7, 0.8]), but the best value is 0.3.

General conclusion

- From the analysis of the graph it emerges that to maximize the popularity of a song, it is advisable to aim for a balance between danceability, energy and acousticness. The graph suggests favoring danceability of 0.3, acousticness of 0.7, and energy of 0.2 . This combination offers the right balance between rhythm, intensity and acoustic elements, increasing the probability that the song will be appreciated by a large audience.

2 MOST USED BPM LEVELS

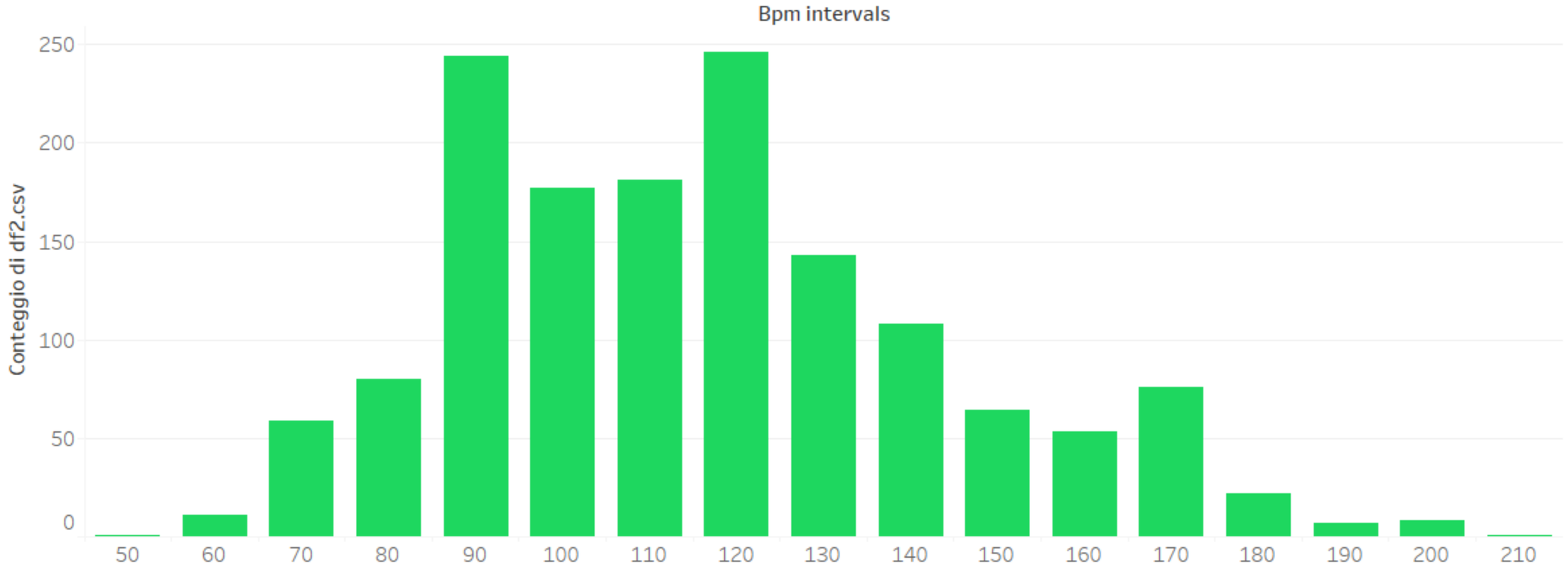


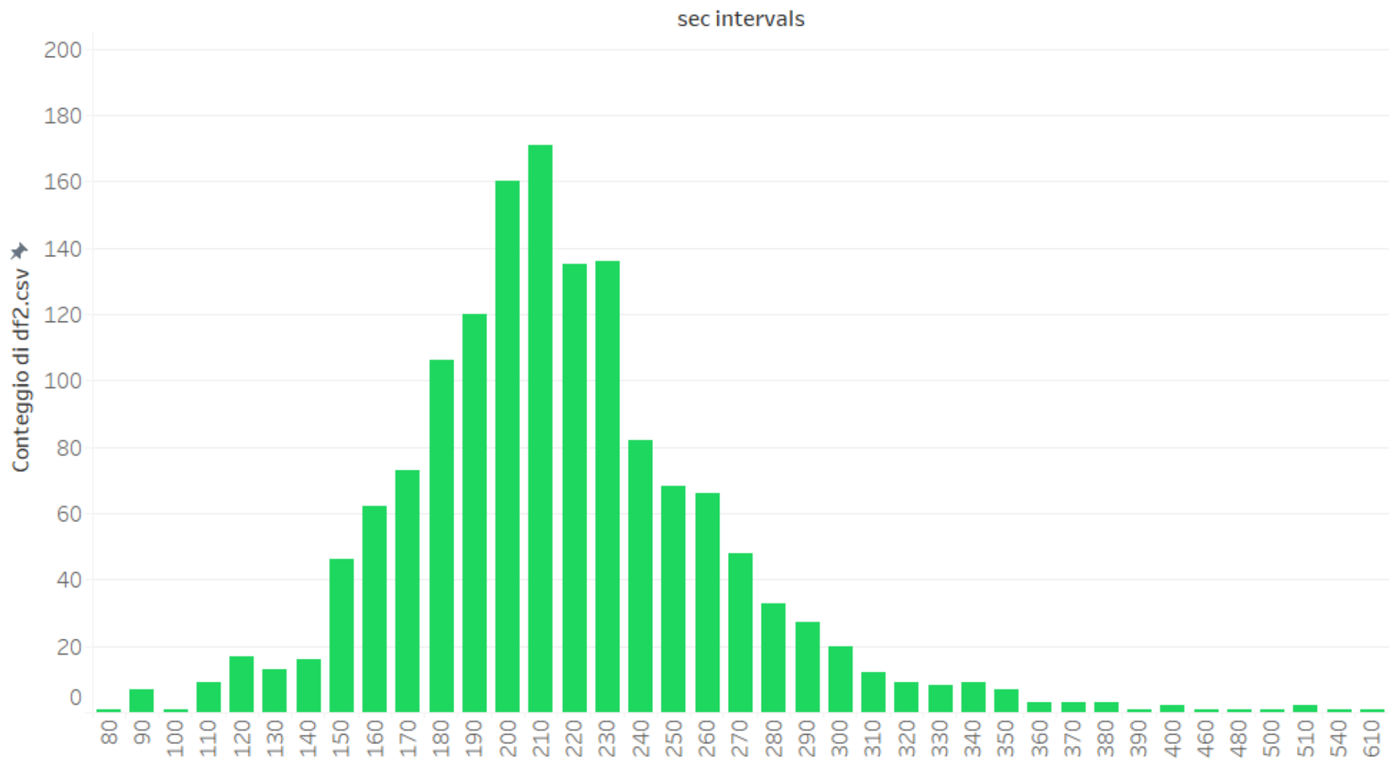
Chart Description:

The chart shows the distribution of the most listened songs on Spotify based on their BPM (beats per minute) level, divided into specific ranges. BPM ranges were created by rounding the BPM value to the nearest ten, making it easier to analyze the distribution of songs based on their tempo.
The vertical axis represents the number of songs for each BPM range, while the horizontal axis indicates the different BPM ranges considered.

Analysis and conclusions:

- The most common BPMs among the most listened songs are concentrated in the 90 and 120 ranges, with over 200 songs in each category. This suggests that successful songs tend to have a moderate tempo, often associated with genres such as pop, hip-hop and dance.
- The 110 and 100 BPM ranges also show a significant presence (more than 150 songs), confirming that a medium tempo is more frequent among the most listened songs.
- Very high or very low BPMs are less frequent. The ranges 50, 60, 190, 200, 210 BPM have a very low number of songs, indicating that extremely slow or very fast tempos are less represented among the most listened to hits.
- In general, most of the most listened to songs have a BPM between 90 and 130, which suggests that a moderate tempo is more appreciated by the public.

3 DURATION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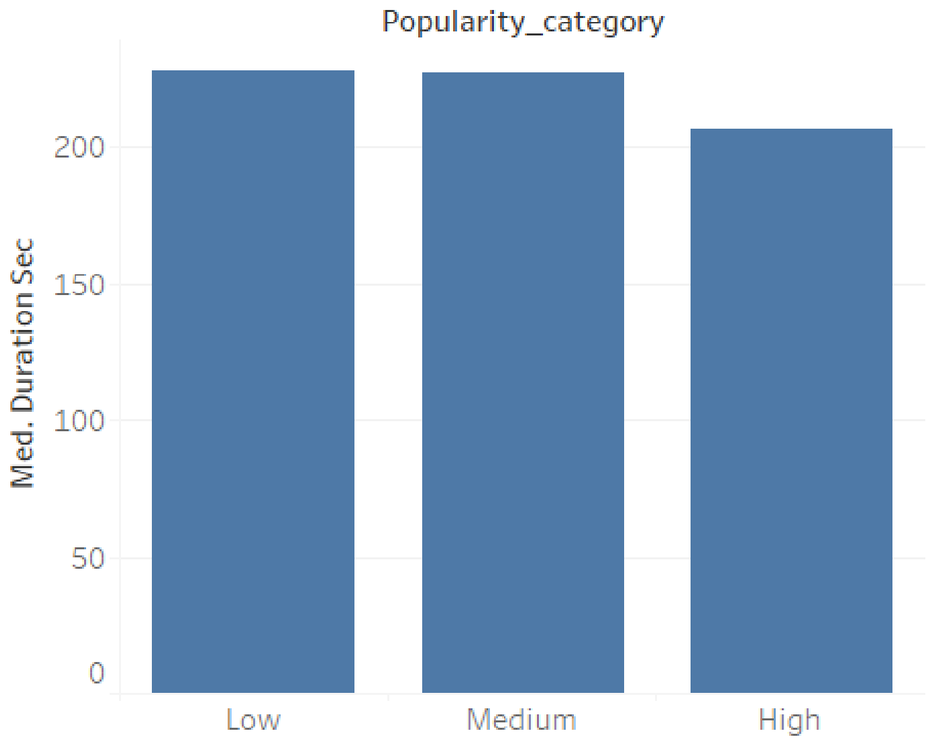

Popularity_category
Med. Duration Sec
200
150
100
50
0
Low
Medium
High

Chart Description:

Left Chart: Distribution of Tracks by Duration

- This histogram shows the distribution of songs by their duration in seconds. The x-axis shows duration intervals in seconds (from 90 to 540 seconds). The y-axis shows the song count.

Right Chart: Distribution of Tracks by Popularity

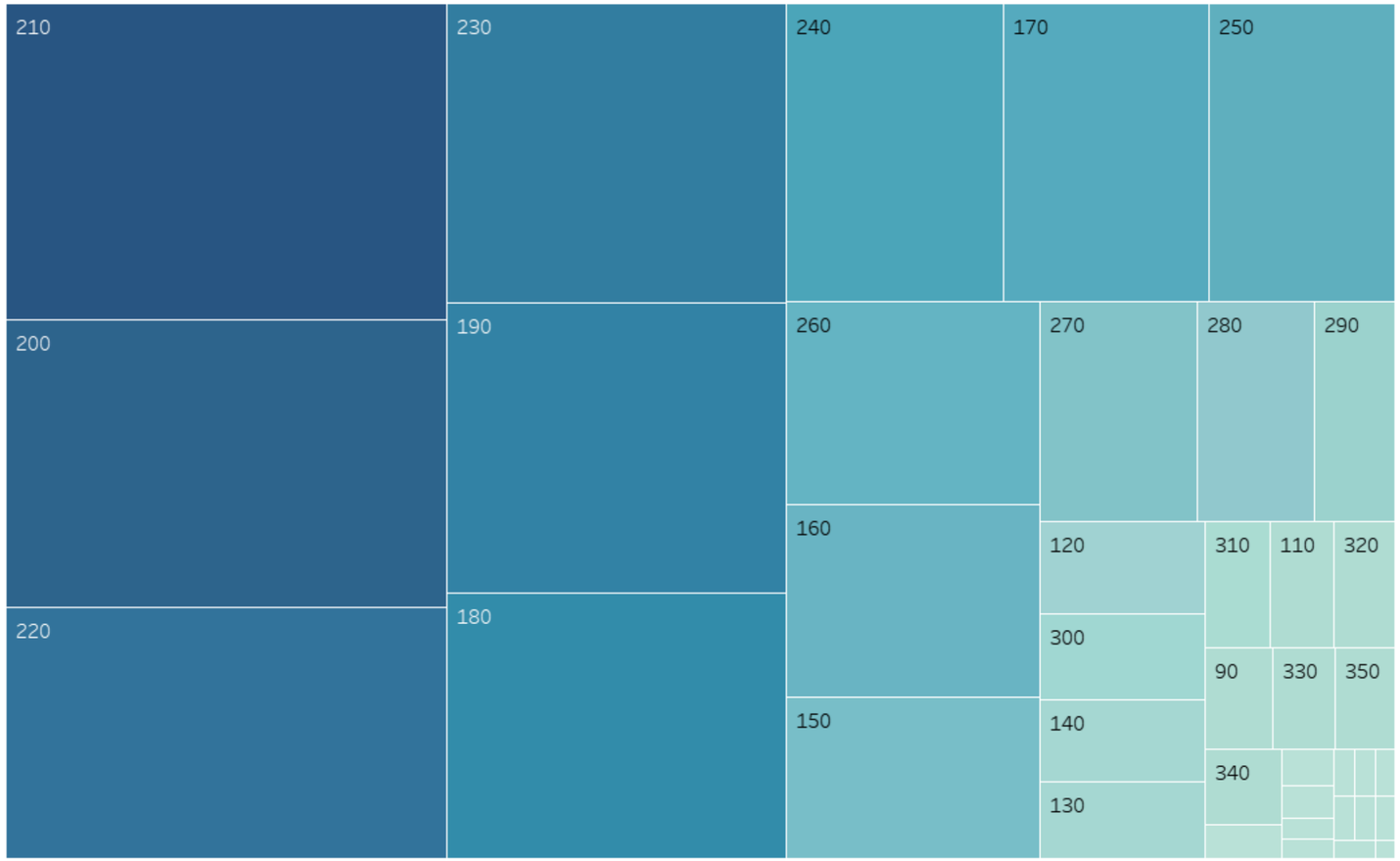
- This bar chart shows the median duration of songs across three popularity categories. The x-axis shows the following popularity categories: "Bassa" (Low), "Media" (Medium), and "Alta" (High)The y-axis shows median duration in seconds.

Analysis and conclusions:

Left chart Distribution of tracks by duration:
The distribution forms a bell curve with a positive skew. Most songs cluster between 170-240 seconds (3-4 minutes). The peak occurs around 200-220 seconds (3:20-3:40 minutes). There's a sharp decline in frequency for songs longer than 300 seconds. Very few songs are shorter than 120 seconds or longer than 360 seconds

Right chart Distribution of tracks by popularity:
Low and Medium popularity songs have nearly identical median durations (around 220 seconds). High popularity songs have slightly shorter median durations (around 205 seconds)

3 DURATION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340
130

Chart Description:

This treemap chart visualizes the relationship between song duration (in seconds) and popularity. Colors and Sizes: Each rectangle represents a specific duration range of songs (in seconds). The size of each rectangle corresponds to the total popularity of songs within that duration range. The color intensity varies according to popularity, with darker shades indicating higher popularity.

Analysis and conclusions:

Songs that are around 210 seconds (3 minutes and 30 seconds) appear to have the highest popularity, with a total score of 11,499. Other durations between 200 and 240 seconds also show significant popularity, suggesting that this may be an optimal song length for success. Shorter songs (e.g., 90–150 seconds) and longer songs (over 300 seconds) tend to have lower popularity. Interpretation: The chart suggests that songs between 3 and 4 minutes in length are generally more successful. This may be influenced by listener preferences and streaming platform algorithms, which often favor songs that fit well within playlists and maintain user engagement.

CONCLUSION

Streaming success on Spotify correlates with specific song characteristics. Leading artists like Taylor Swift and The Weeknd demonstrate that both catalog depth and individual hit impact drive streams.

The most popular tracks feature moderate danceability (0.3), higher acousticness (0.7), and controlled energy levels (0.2). Songs with tempos between 90-130 BPM perform best, with optimal duration around 3:30 minutes.

This data provides a framework for artists and producers to make strategic creative decisions while allowing for artistic expression within these parameters.